

Table 2

0024

Drill Hole Summary

Hole Number	Date Drilled	Approx. Co-ordinates		Hole Depth	Depth to Stratigraphic Units				
		Northing	Easting		Tregolana	Whyalla	Woocalla	Pandurra	
<u>MacFarlane Area :</u>									
MF 1	19/6/73	489,500	650,000	94.5	-	14	29.5	38	
MF 2	21/6/73	488,500	655,000	94.5	-	25	29	71	
MF 3	21/6/73	489,500	653,000	15	-		Not Reached		
MF 4	22/6/73	496,000	657,000	91.5	-	4	20	89	
MF 5	23/6/73	488,000	665,000	96	-	7	16	95	
MF 6	24/6/73	490,000	675,000	105	-	5	99	Not Reached	
<u>Oakden Hills Area :</u>									
OK 1	12/5/73	505,000	700,500	72	4	49	Not Present	70	
OK 2	13/5/73	500,500	699,000	25.5	-	0	Not Present	24	
OK 3	13/5/73	500,000	693,500	47.5	4	5.5	Not Present	43	
OK 4	13/5/73	507,000	694,000	41	-	8	Not Present	39	
OK 5	14/5/73	510,000	693,000	22	-	5	Not Present	21	
OK 6	14/5/73	510,000	690,000	9	-	1	Not Reached		
OK 7	15/5/73	507,500	686,000	85	3	35	Not Reached		
OK 8	15/5/73	499,000	690,000	81	-	0	53	74	
OK 9	26/6/73	496,000	685,000	108	3	34	Not Reached		
OK 10	27/6/73	496,500	703,000	42	2	19.5	26	38.5	
OK 11	29/6/73	512,000	698,500	45	-	15	32	41.5	
OK 12	29/6/73	511,000	699,000	31.5	-	9	Not Present	19.5	

5. EXPLORATION ACTIVITIES

Pacminex activities in the area began with a number of geological reconnaissance traverses towards the end of 1972, before the exploration licence was applied for.

Geological mapping on 1:30,000 scale air photos has been completed over the Birthday-Mt. Moseley-Oakden Hills area. During mapping, rock chip samples were taken for geochemical analysis. This work has indicated an area of anomalous copper values in the vicinity of Little Selby Dam (see geochemical plan).

A small power auger was used to take sub-surface soil samples on a traverse south-west from Birthday Siding. The depth of penetration was only 1 metre, which is considered far too shallow to detect any geochemical anomalies which may be present.

A scout drilling programme was completed in May and June, 1973, using a Mayhew 1000 rotary drilling rig. Twelve holes were drilled in the Oakden area and six in the MacFarlane area. (Drill hole positions are marked on the 1:125,000 regional geology map.) Table 2 gives a summary of the drilling results.

6. PANDURRA FORMATION

6.1 Distribution

Two major facies of the Pandurra Formation are evident in the area. The main outcrops of Pandurra Formation are along the ridge to the west of Pernatty Lagoon, where the rock type is red quartzite, capped in places by silcrete. Smaller outcrops occur in a series from Selby Dam towards Old JRP Dam, with an isolated occurrence to the east of Mt. Moseley.

The second major facies occurs on the western side of Lake MacFarlane, where the unit outcrops as reddish-brown sandstone, partly capped by silcrete.

6.2 Quartzite Facies

Outcrops of this rock are the typical red quartzites seen in the Pernatty Lagoon area - a very tough rock, well bedded, with "liesegang" rings, cross laminations, gritty horizons, and generally strong fracturing.

In drill hole OK 12, 0.5 m of brecciated quartzite was followed by 5 m of white, grey and pink shale, then by 4 m of interbedded quartzite and shale, before massive quartzite was encountered. It may be inferred from this that shale lenses are common through the Pandurra Formation, but, being soft rocks, do not form outcrops.

6.3 Sandstone Facies

The Lake MacFarlane area has not yet been mapped, so knowledge of the sandstone facies is somewhat sketchy. In outcrops to the north of Six Mile Dam (on the western side of the lake) the sandstone consists of quartz, chert and acid volcanic grains, with scattered pebbles of acid volcanics. Grains are sub-angular to sub-rounded, and moderately well sorted. The rock is much less compact than the quartzite facies.

The sandstone facies was intersected in drill holes MF 1 and MF 2. MF 1 intersected 37 m of gritty lithic sandstone, followed by 20 m of fine red sandstone. The gritty sandstone consisted of rounded to sub-angular grains, with noticeable authigenic overgrowths, made up of about 75% quartz and 25% lithic grains (probably acid volcanics). The red sandstone had a similar composition,

but included some fine iron oxide material, probably forming the cement. MF 2 was drilled through 24 m of red coloured sandstone similar to that intersected in MF 1, but generally coarser in grain size.

6.4 Geochemistry

Background geochemical values for Cu, Pb and Zn are quite low : 10-40 ppm Cu, < 20 ppm Pb and 5-40 ppm Zn. No other elements have been determined. At the top of the unit, somewhat higher values are associated with obvious sulphide mineralisation in several drill holes, notably OK 12 (220 ppm Cu).

Outcrops of Pandurra Quartzite south-west of Old JRP Dam gave anomalous Cu geochemical values of up to 120 ppm. To the west of Lake MacFarlane, outcrops of red sandstone gave values up to 340 ppm Cu.

6.5 Post-Pandurra Surface

After lithification, the Pandurra Formation was uplifted and exposed to the agencies of erosion, forming a surface of gentle ridges and broad valleys. This erosional surface is marked by a quartzite breccia with a red clay and sand matrix, similar to that found in the Cattle Grid area. This probably represents a fossil scree slope or soil profile.

Drilling and mapping show that the main Pandurra ridge on the western side of Pernatty Lagoon extends into the Oakden Hills area as a complex of low ridges and valleys (see Structural and Palaeogeographic Map). To the west and south of Oakden Hills, the post-Pandurra surface slopes away to a broad shallow basin of deposition, rising again at Lake MacFarlane in the west, and in the vicinity of Strawbridge Hill to the south. The depth to the top of Pandurra Formation in the basin is as shown in Table 3.

Table 3

Depth of Oakden-MacFarlane Basin

Drill Hole	Depth to Pandurra Formation	Drill Hole	Depth to Pandurra Formation
MF 1	38	Noranda - LD 1	99
MF 2	71	LD 2	190
MF 4	89	LD 3	195
MF 5	95	LD 4	137
MF 6	>105	LD 5	84
Yalymboo Water Bore	187	LD 6	68

7. WOOCALLA DOLOMITE

7.1 Distribution

The Woocalla Dolomite is not continuous over the whole of the area. It fills depressions in the post-Pandurra surface, and forms an on-lap structure onto the Pandurra ridges. Away from the ridges, the dolomite undergoes a facies change to a deeper water black shale facies.

There are no outcrops of the unit in the area, but it was intersected in a number of drill holes. In the northern part of the area, in the vicinity of Old Oakden Hills homestead, the dolomite laps onto the main Pandurra ridge, and was intersected in drill hole OK 11. Holes OK 2 to OK 6 were drilled on an area of Pandurra "highs", and no dolomite was intersected, although chips of dolomite at the base of the Whyalla Sandstone in OK 4 indicate the proximity of dolomite in this locality.

Hole OK 1 intersected no dolomite, even though the depth to Pandurra Formation of 70 m shows that a depression exists in the post-Pandurra surface at this point. OK 10 intersected 12.5 m of dolomitic shale where the unit laps onto the western side of the main Pandurra ridge.

To the west and south of the Oakden area of Pandurra ridges, the deeper water black shale facies was intersected in holes OK 8, MF 2, MF 4, MF 5, MF 6 and in the Noranda holes LD 1 to LD 6. In the MacFarlane area, where the post-Pandurra surface rises again, the black shale facies changes back into the shallow water dolomite facies. 8.5 m of shaley dolomite was intersected in MF 1, on the western shore of Lake MacFarlane.

7.2 Dolomite Facies

The term "dolomite facies" actually encompasses a variety of dolomitic rock types as shown by mapping in the Pernatty Lagoon and Woocalla areas. However, in the Oakden Hills-Lake MacFarlane area, lack of outcrop prevents the mapping of facies changes in the dolomite.

A number of variations are evident in the drill cuttings. OK 11 intersected 3.5 m of tough white to pale brown dolomite overlain by 6 m of dark grey dolomitic mud. OK 10 intersected 2 m of tough white to pale grey dolomite overlain by 10.5 m of dark grey dolomitic shale. MF 1 was drilled through 8 m of finely laminated grey to brown shaley dolomite.

The dolomite also shows a considerable variation in trace metal content. Table 4 shows the maximum values for Cu, Pb and Zn recorded in the dolomite in the various drill holes.